Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | sahaj-bhadja | sahajbhadja | sahaj-bhadja

Summary

Product software engineer with 2+ years building scalable data-processing applications, batch pipelines, and near-real-time services. Skilled in Python, Java, SQL, AWS, Databricks, Docker, automated testing, performance optimization, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant data services on Azure SQL and Blob Storage for 500+ users, implementing normalized models, validation, and indexed queries to reduce analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant batch workflows for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and production monitoring, improving throughput by 3x while preventing cascading pipeline failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered production Python services and REST integrations handling 10,000+ daily requests, combining secure ingestion, AWS RDS persistence, and AI-assisted processing while maintaining stable performance during peak usage.
- Implemented concurrent data processing for 1,000+ files with validation, error isolation, structured logging, code reviews, and automated tests, reducing processing turnaround by 60% for 150+ users.

USC AD Labs **Los Angeles, CA**
Machine Learning Researcher *January 2025 – May 2025*
- Built Python data and simulation pipelines processing 100,000+ trajectory logs, improving long-horizon planning by 18% through reproducible evaluation, structured experiment tracking, performance analysis, and automated visualization.
- Architected a spatial-temporal model combining HD-map geometry and agent trajectories through multi-head attention, improving closed-loop stability by 15% while generating explainable visual analysis of production system behavior.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform microservices with Jenkins pipelines, automated VM builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening integration quality and operational production reliability.
- Optimized ETL and API workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving response times by 30% and data-pipeline efficiency by 68% for downstream consumers.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for booking, scheduling, payment, and status workflows across 50,000+ records, integrating Firebase event persistence with real-time synchronization, structured validation, and responsive cross-platform application clients.
- Developed Databricks and Spark micro-batches for booking and payment events, reducing manual handling by 70% while improving analytical processing, downstream data consistency, and confirmation speed by 40%.

Skills

Languages & Data: Python, Java, SQL, JavaScript, TypeScript, Bash, data structures, data modeling
Data Engineering: batch processing, near-real-time pipelines, ETL, Apache Spark, Databricks, AWS Glue, data validation
Cloud & Infrastructure: AWS, Azure, GCP, Amazon S3, AWS RDS, Docker, Kubernetes, Linux, CI/CD
Databases & Platforms: PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB, Firebase, Snowflake, REST APIs
Testing & Operations: PyTest, Mockito, unit testing, integration testing, debugging, profiling, monitoring, production support
Engineering Practices: Git, Jenkins, code reviews, Agile, requirements analysis, technical documentation, cross-functional collaboration, AI-assisted development

Projects

Chat2SQL | *Java, TypeScript, SQL, Snowflake, Docker, Maven*
- Built a full-stack data workspace supporting 10+ relational and NoSQL systems with schema exploration, validated query execution, AI-assisted SQL generation, and cross-platform deployment across Java and TypeScript modules.

Linguistix | *Python, AWS, Docker, CodePipeline, REST APIs*
- Developed a containerized data-processing service with automated releases, validation checks, structured error handling, and scalable REST interfaces, achieving 87% accuracy while strengthening repeatable deployment and production reliability.

FinNexus | *TypeScript, Kubernetes, MongoDB, WebSockets, Plaid*
- Built secure near-real-time integrations supporting 50,000+ users and 1M+ transactions, combining token-based APIs, persistent filtering, WebSocket synchronization, and Kubernetes-backed deployment automation across high-volume production data workflows.